Hence a committee of 5 members (2 men and 3 women) is selected in

$6C_2 \times 5C_3$ ways

i.e., $\frac{6 \times 5}{2 \times 1} \times \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 150$ ways.

Example 7.11

How many triangles can be formed by joining the vertices of a pentagon of five sides.

Solution:

There are 5 vertices in a pentagon. One triangle is formed by selecting a group of 3 vertices from given 5 vertices. This can be done in $5C_3$ ways.

i.e., Number of triangles $= 5C_3 = \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 10$

Example 7.12

A question paper contains section A with 5 questions and section B with 7 questions. A student is required to attempt 8 questions in all, selecting at least 3 from each section. In how many ways can a student select the questions?

Solution:

Selection of 8 questions from 12 questions and at least 3 from each section is given below

Section A 5Questions	Section B 7Questions	Combinations	NO. of ways
3	5	$5C_3 \times 7C_5$	210
4	4	$5C_4 \times 7C_4$	175
5	3	$5C_5 \times 7C_3$	35
		Total	420

Therefore total number of selection is 420

Example 7.13

If $6P_r = 360$ and $6C_r = 15$ find r.

Solution:

From the formula,

$$nC_r = \frac{nP_r}{r!}$$

$$6C_r = \frac{6p_r}{r!}$$